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TECHNOLOGICAL UNIVERSITY DUBLIN

CITY CAMPUS

TU256 Postgraduate Certificate in Data Science

TU059/TU060 MSc in Computer Science

Year 1 and 2

SEMESTER 2 EXAMINATIONS 2021/22

Machine Learning

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ANSWER 2 OF 3 QUESTIONS

1. (a)
 - i) Machine learning prediction models need to be *consistent* with the training data and to be able to *generalise* well. Explain what both these terms mean [4 marks]
 - ii) Explain what *overfitting* is in supervised machine learning and describe one way of identifying it. [4 marks]
 - iii) When finding nearest neighbours, which distance functions would you use when comparing examples with (a) numeric features and (b) categorical features. [4 marks]
 - iv) A limitation of some clustering approaches is that they can be *non-deterministic*. Explain what this means and why it happens. [4 marks]
 - v) Data leakage can give over-optimistic model performance. Explain why and discuss what is necessary to avoid data leakage. [4 marks]
 - vi) Explain why k-fold cross validation provides a more robust evaluation in a classification problem when compared with a single random train-test split. [4 marks]
- (b) Compare and contrast decision trees and nearest-neighbour classifiers as mechanisms for developing supervised learning systems. [5 marks]
- (c) The table below shows a dataset of 8 examples described by 3 categorical features, representing booking decisions made by hotel customers. Each example has one of two class labels: **Book?** = {yes, no}.

Customer	Stars	Pool	Gym	Book?
x1	3	Y	Y	yes
x2	2	N	N	no
x3	3	N	Y	no
x4	2	Y	N	no
x5	4	Y	Y	yes
x6	3	N	N	no
x7	2	N	Y	no
x8	4	Y	N	yes

- i) Calculate the overall entropy for the dataset showing the different elements in the calculation. [6 marks]

Q1 continues on the next page

- ii) Using Information Gain, find the best feature to split the root of a Decision Tree classifier built on the training data above. Show all of your calculations. [15 marks]

2. (a) i) Explain, with an example, why *smoothing* is necessary in Naïve Bayes classifiers. [5 marks]

- ii) The dataset below contains observations for cases of sunburn and two cases Q1 and Q2 for which the outcome is unknown. Using a Naïve Bayes classifier determine the most likely outcome for both Q1 and Q2. Clearly describe the steps you take, any assumptions you make and explain all calculations and probabilities that you use. [20 marks]

Gender	Hair	Height(cms)	Age	City/Rural	Outcome
female	blonde	160	28	rural	sunburned
male	blonde	182	15	city	none
female	brown	150	45	city	none
female	brown	152	59	rural	sunburned
female	red	125	10	rural	sunburned
male	brown	190	25	rural	none
male	brown	158	35	rural	none
female	brown	135	18	city	none
male	red	162	42	rural	sunburned
male	blonde	188	56	city	none

Q1:

female	blonde	136	50	city	none
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Q2:

male	red	185	16	rural	none
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- (b) i) Give three reasons to reduce the number of features in a dataset. [6 marks]
- ii) Explain the difference between *filters* and *wrappers* for feature subset selection, with reference to an example of each type of approach. [4 marks]
- iii) Why might filters and wrappers select different feature subsets when applied to the same dataset? [4 marks]

Q2 continues on the next page

- (c) i) The confusion matrix below summarises the performance of a binary spam email classifier. Indicate the performance measure(s) you would use to evaluate this classifier. Discuss whether you think this classifier is performing well on the task. [6 marks]

Predicted Class		
Spam	Non-Spam	
220	55	Spam
50	170	Non-Spam

Actual Class

- ii) Explain, using an example, why classification accuracy might not always be an adequate measure of predictive performance. [5 marks]
3. (a) i) *Gradient descent* is a key component of error-based machine learning algorithms. Explain gradient descent and show — with the aid of a diagram — how it can be used to optimise the parameters of a machine learning model. [7 marks]
- ii) Explain the function of the learning rate parameter (α) in gradient descent. State the potential problems associated with setting the learning rate too high and too low. [6 marks]
- (b) A k -Nearest Neighbours classifier has been trained on the following data to predict whether the weather is suitable for going outside (as indicated by the **Go Out?**) column. The classifier uses Manhattan distance with the hyper-parameter k set to 1. The general formula for Minkowski distance is provided in the supplemental material.

Q3 continues on the next page

Id	Temp	Humidity	Windspeed	Cloudy?	Go Out?
1	18	22	24	yes	no
2	22	10	10	no	yes
3	24	16	12	no	yes
4	14	28	10	yes	no
5	18	14	14	yes	yes
6	12	16	18	no	yes
7	20	4	8	yes	yes

- i) What prediction will the model make on the following query? [12 marks]
Show your workings.

Temp	Humidity	Windspeed	Cloudy?	Go Out?
14	20	10	yes	?

- ii) Comment on the suitability of setting $k = 2$ for this model. [3 marks]
- iii) Comment on the suitability of setting $k = 3$ for this model. [3 marks]
- iv) The model described above uses un-normalized data. Explain the impact this will have on the model. [3 marks]
- (c) i) Describe, with the aid of a diagram, the steps involved in building an evaluation pipeline using hold-out testing to compare the performance of two competing models. [3 marks]
- ii) When building an evaluation pipeline, what precautions should be taken to avoid data leakage? [4 marks]
- iii) Grid search is often a key component of hyper-parameter selection in an evaluation pipeline. Describe the benefits and drawbacks of employing grid search as part of the model building process. [6 marks]
- iv) Describe an alternative approach to grid search which addresses the drawbacks. [3 marks]

Supplemental Material

Useful Formulae

Entropy of a dataset of examples D with labels $t_1, t_2, \dots, t_l$

$$H(D) = - \sum_{i=1}^l (P(t_i) \times \log_2(P(t_i)))$$

Information Gain for descriptive feature d that splits D into partitions $D_1, D_2, \dots, D_k$

$$IG(d, D) = H(D) - \sum_i^k \frac{|D_i|}{|D|} \times H(D_i)$$

Minkowski Distance formula

$$Minkowski(\mathbf{a}, \mathbf{b}) = \left(\sum_{i=1}^m abs(a[i] - b[i])^p \right)^{\frac{1}{p}}$$

Table of base 2 log for different fractions

$\log_2(a/b)$		a													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
b	1	0.00													
	2	-1.00	0.00												
	3	-1.58	-0.58	0.00											
	4	-2.00	-1.00	-0.42	0.00										
	5	-2.32	-1.32	-0.74	-0.32	0.00									
	6	-2.58	-1.58	-1.00	-0.58	-0.26	0.00								
	7	-2.81	-1.81	-1.22	-0.81	-0.49	-0.22	0.00							
	8	-3.00	-2.00	-1.42	-1.00	-0.68	-0.42	-0.19	0.00						
	9	-3.17	-2.17	-1.58	-1.17	-0.85	-0.58	-0.36	-0.17	0.00					
	10	-3.32	-2.32	-1.74	-1.32	-1.00	-0.74	-0.51	-0.32	-0.15	0.00				
	11	-3.46	-2.46	-1.87	-1.46	-1.14	-0.87	-0.65	-0.46	-0.29	-0.14	0.00			
	12	-3.58	-2.58	-2.00	-1.58	-1.26	-1.00	-0.78	-0.58	-0.42	-0.26	-0.13	0.00		
	13	-3.70	-2.70	-2.12	-1.70	-1.38	-1.12	-0.89	-0.70	-0.53	-0.38	-0.24	-0.12	0.00	
	14	-3.81	-2.81	-2.22	-1.81	-1.49	-1.22	-1.00	-0.81	-0.64	-0.49	-0.35	-0.22	-0.11	0.00